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Serum metabolomic profiles linking diabetes mellitus to pulmonary hypertension: a prospective cohort study with mediation analysis and risk prediction

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Abstract

Background Metabolic dysregulation in patients with diabetes mellitus (DM) may contribute to pulmonary vascular remodeling, yet biomarkers for pulmonary hypertension (PH) risk in diabetes remain unclear.

Objectives To assess the association between DM and incident PH, quantify mediation by circulating metabolites, and develop a metabolite-based PH risk model in diabetes.

Methods In a prospective cohort of 459,714 UK Biobank participants without PH at baseline, Cox proportional hazard models estimated the association between DM and incident PH. Nuclear magnetic resonance (NMR) metabolomics, mediation analysis, and LASSO regression were used to identify metabolic mediators and construct a predictive model.

Results During a median follow-up of 13.25 years, 2,250 PH cases occurred. DM was associated with a 39.6% higher PH risk (HR = 1.396, 95%CI: 1.255–1.553; $P < 0.001$), and earlier DM onset was associated with stronger risk ($P < 0.001$). Seven NMR-derived metabolites—related to hyperglycemia, inflammation (glycoprotein acetyls), amino acid and lipid metabolism—mediated 37.26% of the total effect. Including these metabolites improved prediction in diabetes (C index from 0.692 to 0.735; $P < 0.001$).

Conclusions DM is an independent risk factor for PH. Integration of NMR-derived metabolites improves risk prediction and suggests hyperglycemia and inflammation as key mechanisms, supporting the role of metabolomics in PH stratification and prevention among diabetes.

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Keywords Pulmonary hypertension, Diabetes mellitus, Metabolomics, Mediation analysis, Risk prediction

Introduction

Pulmonary hypertension (PH) is a progressive disorder characterized by elevated pulmonary arterial pressure and vascular remodeling, leading to right heart failure and premature death [1]. In a hemodynamic evaluation of 261 PH patients across Duke University and Cleveland Clinic, 21% had diabetes mellitus (DM), and these patients exhibited significantly worse survival outcomes [2]. However, large-scale, prospective data evaluating the association between DM and incident PH remain limited. DM is a common metabolic disorder that has been implicated in the pathogenesis of PH through mechanisms including insulin resistance, and endothelial dysfunction [3]. It is well established that DM increases the risk of cardiovascular disease, nephropathy, and neuropathy-complications that are partly driven by altered lipid and amino acid metabolism, oxidative stress, and mitochondrial dysfunction [4, 5]. Notably, several metabolites altered in DM have also been found aberrantly expressed in PH patient-derived cardiopulmonary tissues, cellular and animal models of PH [6]. This emerging evidence implies that metabolic dysregulation may serve as a key intermediate step in DM-associated PH, but the metabolic pathways remain poorly defined.

To bridge these gaps, we integrated a large-scale cohort study from the UK Biobank, integrating nuclear magnetic resonance (NMR) metabolomics to characterize the metabolic profile associated with DM and its potential role in PH development. The advent metabolomics allows for high-throughput, highly reproducible quantification of a wide range of circulating metabolites across interconnected biochemical pathways, offering distinct advantages over conventional methods [7–9]. We aimed to (1) evaluate the prospective association between DM and PH in a large population-based cohort; (2) identify circulating metabolites mediating this association; and (3) assess whether incorporating metabolomic signatures improves PH risk prediction in individuals with DM.

Methods

Study population

UK Biobank is a population-based prospective cohort study of over 500,000 participants aged 40–69 years across the UK, initiated in 2006 [10]. The collected data and samples will be correlated with participants' existing health records to allow for longitudinal follow-up of disease morbidity and mortality [11]. UK Biobank received ethical approval from the Northwest Multicenter Research Ethics Committee, and all participants signed informed consent forms.

In the present study, we excluded participants with pre-existing PH at baseline ($n=123$), those with DM diagnosed subsequent to PH ($n=278$), and individuals with missing data on covariates ($n=42,241$), leaving 459,714 participants eligible for inclusion (Fig. 1). To explore the role of the NMR metabolites as mediators in the association between DM and PH, additional exclusions were made for individuals lacking metabolite data, resulting in a final metabolic subgroup of 237,848 participants. The sample size was further reduced to 42,270 individuals diagnosed with DM, allowing for a more focused investigation of the potential impact of age at DM onset on outcomes. Finally, predictive models for NMR metabolites were constructed using data from a cohort of 23,478 individuals diagnosed with DM.

NMR-Based plasma metabolite profiling

A total of over 118,000 baseline non-fasting plasma samples were evaluated for metabolites using a high-throughput and validated NMR metabolomics platform provided by Nightingale Health. The metabolomic profiling encompassed a comprehensive array of quantitative molecular data, including 14 lipoprotein particles subclasses, from small high-density lipoprotein particles (S-HDL-P) to chylomicrons and extremely large very-low-density lipoprotein particles (XXL-VLDL-P), along with their lipid concentrations and composition [8]. Additionally, detailed metrics of cholesterol and triglyceride, apolipoprotein A1 and B, fatty acid compositions, albumin, and various low-molecular-weight metabolites such as amino acids, ketones, and glycolysis-related metabolites were evaluated. This metabolomic panel has been widely employed in epidemiological studies [12–14].

Plasma metabolites were quantified using Nightingale Health's high-throughput ^1H -NMR platform as part of the UK Biobank biomarker enhancement initiative. EDTA plasma samples from aliquot 3 were prepared using standardized automated liquid-handling procedures and measured in Nightingale Health's ISO 13,485-accredited laboratory between 2019 and 2020. Two complementary NMR spectra (presaturated and T_2 -filtered) were acquired on 500-MHz Bruker instruments, and 249 metabolic biomarkers (168 absolute concentrations and 81 ratios) were derived using Nightingale's validated quantification library [8]. Analytical validity was ensured through a rigorous quality-management framework: each 96-well plate included internal controls and UK Biobank blind duplicates, enabling continuous monitoring across six spectrometers. As reported in the UK Biobank Phase 1 QC summary, most biomarkers exhibited intra- and

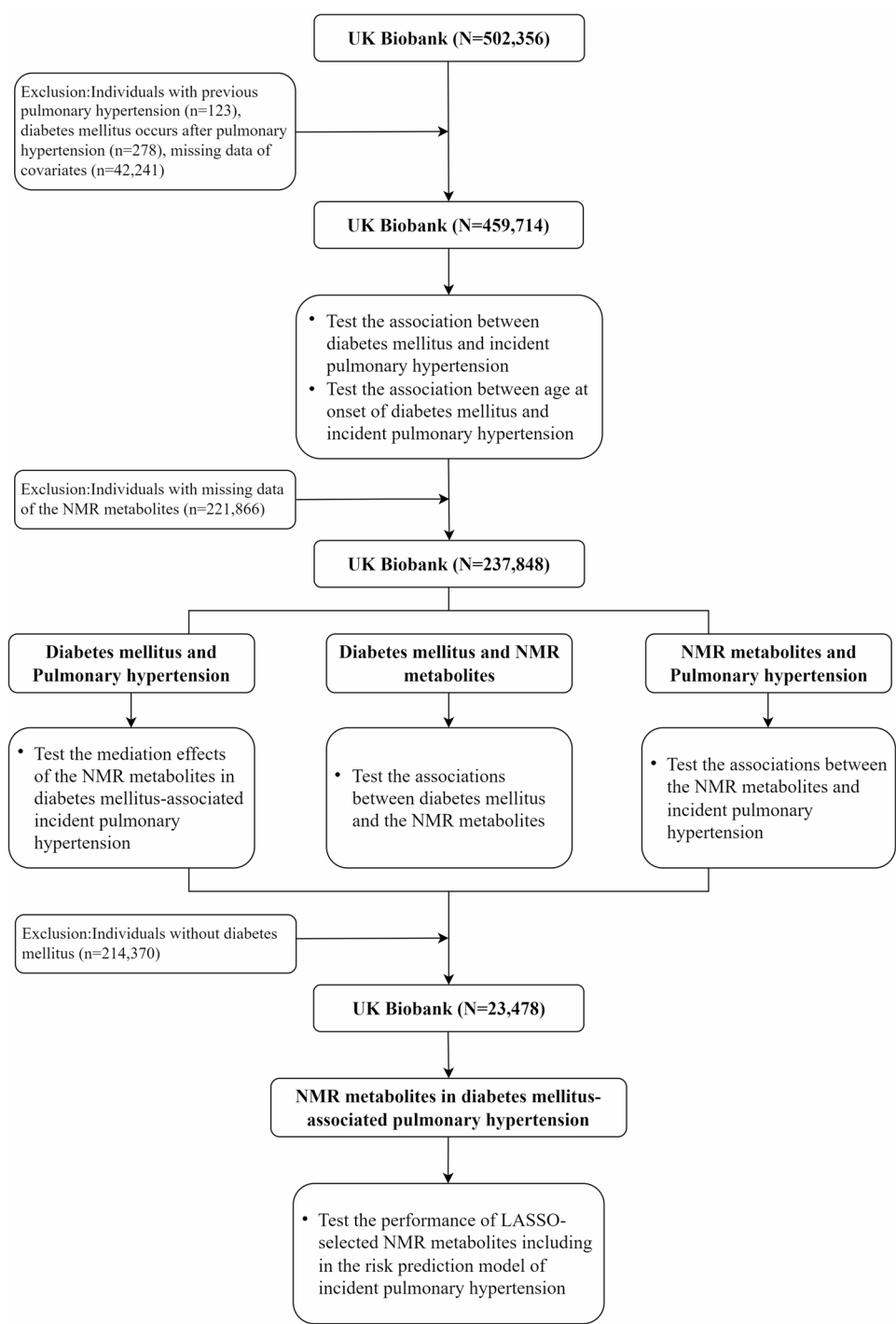


Fig. 1 Flow chart of the study design and workflow

inter-assay coefficients of variation < 5%, and nearly all remained < 10%, indicating excellent reproducibility and minimal batch effects. QC flags were used to identify potential interference, and sensitivity analyses excluding flagged values produced consistent results. For epidemiological analyses, we selected 143 metabolites prioritizing

biomarkers measured directly rather than derived wholly from other variables [15].

Assessment of DM

Cases of DM were identified by (1) hospital-diagnosed DM via ICD-10 codes (E10-E14) from all diagnostic fields in the inpatient records, type 2 diabetes mellitus (T2DM)

was defined by code E11; (2) clinician-confirmed DM through standardized self-reporting, and (3) biochemical validation according to American Diabetes Association criteria (2024) with a fasting blood glucose (FBG) level ≥ 7.0 mmol/L (126 mg/dL) or a glycosylated hemoglobin (HbA1c) level ≥ 48 mmol/mol (6.5%) [16]. The presence of any one criterion is sufficient for diagnosis.

Definition of outcome and follow-up time

The primary outcome of this study was incident PH, which was defined by primary pulmonary hypertension (I27.0) and other secondary pulmonary hypertension (I27.2) [17]. The study utilized hospital admission data from England up to 31 October 2022, from Scotland up to 31 August 2022, and from Wales up to 31 May 2022. Person-years were calculated from the date of enrollment to the date of first incident PH, death, loss to follow-up, or end of follow-up, whichever event occurred first.

Covariates

A range of covariates were incorporated into the analytical model to account for their potential effects. Information on age, sex (male and female), race (White, Black, Asian, and Mixed), smoking status (never, past, and current smokers), alcohol intake status (never, past, and current drinkers), assessment center (England, Scotland, and Wales), household income (<£18,000, £18,000–£30,999, £31,000–£51,999, £52,000–£100,000, and >£100,000), level of education (College/university level degree, Non-College/university level degree, and other), and medication history was collected at baseline by questionnaire or verbal interview. Nurses took body measurements and collected data on height, weight, Systolic blood pressure (SBP) and Diastolic blood pressure (DBP) at baseline. Body mass index (BMI) is quantified in units of kg/m² and is calculated by dividing weight in kilograms by the square of height in meters. Physical activity (PA) was quantified by computing the total metabolic equivalent (MET) minutes per week for all activities, encompassing a range of activities such as walking, moderate-intensity, and vigorous-intensity exercise. According to public health guidelines, participants were categorized as “inactive” if their PA was < 600 MET-minutes/week and “active” if ≥ 600 MET-minutes/week [18, 19].

Statistical analysis

Categorical variables describing baseline characteristics are summarized as number and percentages. Continuous variables are expressed as mean $\pm$ standard deviation (SD) for normally distributed data or median (interquartile range, IQR) for non-normally distributed data. Differences between the observation group (with DM) and the control group (without DM) were assessed using

chi-square tests for categorical variables and independent t-tests or Mann-Whitney U tests for continuous variables, as appropriate.

To assess the association between DM and incident PH, multivariate Cox regression model was employed, with follow-up time as the time scale. Hazard ratios (HR) and 95% confidence intervals (CI) were estimated across three progressively adjusted models: Model 1, adjustments were made for age, sex, and self-identified race; Model 2, further adjusted for BMI, smoking status, and alcohol intake status; Model 3, additionally adjusted for blood pressure, physical activity, assessment center, household income, and educational level. To explore the impact of age at DM diagnosis, age was analyzed both as a continuous variable (per 10-year increment) and as categorical subgroups (<50, 50–60, 60–70, and ≥ 70 years). Multivariate Cox regression models were used to estimate HRs and 95% CIs for PH risk across these age categories.

To investigate the potential the mediating role of metabolites in the association between DM and PH, all metabolites were log-transformed and standardized (mean = 0, SD = 1) prior to analysis. Spearman correlation coefficients were calculated to assess inter-metabolite correlations. Multiple linear regression models were used to examine associations between DM and metabolite levels, while Cox proportional hazards models were employed to assess associations between metabolite level and PH risk. The Benjamin-Hochberg method was used to control for false discovery rate (FDR) to correct for multiple hypothesis testing, and $P_{\text{FDR}} < 0.05$ was considered statistically significant [20]. Mediation analysis was performed to estimate the mediator proportion, total effect, direct effect, indirect effect, and 95%CI for each metabolite and metabolite score. The metabolite score was calculated on the basis of the following weighted average equation: Metabolite Score = $\sum_{i=1}^n \omega_i \bullet x_i$ n is the total number of metabolites, x_i is the z-score standardized to the concentration of the i th metabolite, w_i is the weight value of the i th metabolite.

In a population with confirmed DM, the survival analysis of metabolite expression data was performed using LASSO-regularized Cox regression with 10-fold cross-validation [21]. Metabolites with non-zero coefficients ($P < 0.05$) were incorporated into the Cox proportional hazards model to construct a predictive model for PH risk. Model performance was evaluated using Harrell's C-statistic [22]. The IDI and NRI were calculated with the aim of assessing the additional predictive power of metabolites [23, 24].

Stratified analyses were performed across key covariates. Multiplicative interaction terms were included in the models to test for statistical interactions. In addition, several sensitivity analyses were performed to ensure

the robustness of the findings: Firstly, participants with PH occurring within two years of recruitment were excluded from the study ($n = 2,253$) in order to reduce the possibility of reverse causality. Secondly, exclusion of individuals with conditions known to cause secondary PH at baseline, including left heart failure (LHF), chronic obstructive pulmonary disease (COPD), pulmonary embolism (PE), and interstitial lung disease (ILD) [25]. Thirdly, Multiple imputation by chained equations (MICE) to address missing multivariate data and evaluate the impact of missing values [26]. Fourthly, a supplementary adjustment for the utilization of antidiabetic drugs, antihypertensive drugs, and lipid-lowering drugs. Finally, the analysis was restricted to participants diagnosed with T2DM.

Results

Baseline characteristics

The mean age of participants was 56.52 years, and 54.4% were female. At baseline, 46,128 participants (10%) had DM. During a median follow-up of 13.25 years, 2,250 incident PH cases were identified (0.5% incidence). Compared to those without PH, PH cases were more likely to be older, male, physically inactive, current or former smokers, and exhibited lower household income and education levels (all $P < 0.001$) (Table 1).

DM confers an increased risk of developing PH

Multivariable Cox regression revealed that DM was significantly associated with incident PH. The diabetic group exhibited a significantly higher PH prevalence rate (0.878 per 1,000 person-years) compared to the non-diabetic group (0.315 per 1,000 person-years), corresponding to a 2.79-fold elevated unadjusted risk (Table 2). After adjusting for age, sex, and ethnicity, Model 1 showed that individuals with DM had a 119.1% increased risk (HR: 2.191, 95% CI: 1.983–2.420, $P < 0.001$). Model 2, further adjusted for BMI, smoking status, and alcohol intake, demonstrated a 53.8% increased risk (HR: 1.538, 95% CI: 1.384–1.710, $P < 0.001$). In the fully adjusted Model 3 still demonstrated a significant 39.6% increased risk (HR: 1.396, 95% CI: 1.255–1.553, $P < 0.001$) (Table 2).

Importantly, stratification by age of DM onset revealed an inverse dose–response relationship. Individuals diagnosed with DM at ≥ 70 years serve as the reference group (HR = 1.0). Compared to the reference group, those with onset between 60 and 70, 50–60, and ≤ 50 years had HRs of 2.106 (95% CI: 1.696–2.615), 3.199 (95% CI: 2.371–4.315), and 6.210 (95% CI: 3.702–10.417), respectively. And every 10-year increase in onset age reducing PH risk by approximately 48.8%, indicating that earlier-onset DM confers a substantially higher PH risk (Fig. 2, Table S1).

These results confirm that DM is a significant and independent risk factor for PH, with earlier onset conferring a more pronounced risk.

DM is associated with widespread metabolic disturbances

To explore potential metabolic mechanisms linking DM to PH, we used NMR-based metabolomics to evaluate systemic metabolic alterations associated with DM. Linear regression showed that 137 of 143 metabolites were significantly dysregulated in diabetic participants ($P_{\text{FDR}} < 0.05$), with core features of lipoprotein remodeling, imbalances in amino acid metabolism and disturbances in energy metabolism. Specifically, diabetic patients exhibited markedly increased VLDL particle mean diameter ($\beta = 0.193$, $P_{\text{FDR}} = 2.37 \times 10^{-198}$), together with significantly lower levels of omega-3 fatty acids ($\beta = -0.051$, $P_{\text{FDR}} = 1.54 \times 10^{-13}$). Some studies suggest this is associated with the liver overproducing large VLDL particles, which carry more triglycerides than small VLDL and correlate with the severity of insulin resistance [27, 28]. Amino acid profiling revealed elevated levels of branched-chain amino acids (BCAAs) (e.g., leucine: $\beta = 0.186$, $P_{\text{FDR}} = 7.39 \times 10^{-167}$), consistent with the known links between amino-acid metabolism, insulin resistance, and cardio-metabolic health [29]. In addition, elevated ketone body metabolites (e.g., 3-hydroxybutyrate: $\beta = 0.075$, $P_{\text{FDR}} = 1.30 \times 10^{-25}$), glucose ($\beta = 1.017$, $P_{\text{FDR}} = 0$) and Glycoprotein Acetyls (GlycA) ($\beta = 0.151$, $P_{\text{FDR}} = 2.11 \times 10^{-114}$) pointed to the combined influence of enhanced lipolysis, impaired insulin signaling, and chronic systemic inflammation (Fig. 3A, Table S2).

Specific Circulating metabolites are significantly associated with PH risk

We then used Cox proportional risk modelling to evaluate the prospective associations between circulating metabolites and PH incidence. Of the 143 metabolites tested, 117 were significantly associated with PH risk after FDR correction ($P_{\text{FDR}} < 0.05$). Panels A and B are aligned by metabolite to facilitate identification of metabolites that are (a) elevated in DM and associated with higher PH risk (glucose, GlycA, 3-hydroxybutyrate), (b) elevated in DM but associated with lower PH risk (leucine, VLDL particle mean diameter, omega-3 fatty acids) or (c) decreased in DM and associated with higher PH risk (creatinine) (Table S2, Table S3).

Protective associations were observed for VLDL particle mean diameter (HR = 0.788, $P_{\text{FDR}} = 2.24 \times 10^{-13}$) and omega-3 fatty acids (HR = 0.770, $P_{\text{FDR}} = 1.89 \times 10^{-18}$). Regarding amino acids, most also showed inverse associations with PH, particularly leucine (HR = 0.840, $P_{\text{FDR}} = 5.02 \times 10^{-8}$). To our knowledge, no prior studies have reported associations between BCAAs and PH, which suggests the existence of previously unrecognized

Table 1 Demographic and diabetic characteristics of participants included in this study (n=459,714)

Characteristics	Overall(n=459,714)	Pulmonary hypertension		P ^a
		No (n=457,464)	Yes (n=2,250)	
Follow-up time, years, Mean±SD	13.25 ±1.95	13.27±1.92	9.55±3.39	<0.001
Sex, n (%)				<0.001
Male	209,490 (45.6%)	208,313 (45.5%)	1,177 (52.3%)	
Female	250,224 (54.4%)	249,151 (54.5%)	1,073 (47.7%)	
Age, years, Mean±SD	56.52±8.09	56.50±8.09	61.91±6.39	<0.001
BMI, Kg/m ² , Mean±SD	27.40±4.77	27.39±4.76	29.62±6.32	<0.001
Physical activity, METS-min/week, n (%)				<0.001
Active	340,472 (74.1%)	338,988 (74.1%)	1,484 (66.0%)	
Inactive	119,242 (25.9%)	118,476 (25.9%)	766 (34.0%)	
DBP, mmHg, Mean±SD	82.21±10.69	82.22±10.69	81.58±11.59	0.005
SBP, mmHg, Mean±SD	139.73±19.67	139.70±19.66	144.74±20.89	<0.001
Ethnic, n(%)				0.001
White	415,996 (90.5%)	413,942 (90.5%)	2,054 (91.3%)	
Black	2,645 (0.6%)	2,633 (0.6%)	12 (0.5%)	
Asian	16,287 (3.5%)	16,243 (3.6%)	44 (2.0%)	
Mixed	16,981 (3.7%)	16,883 (3.7%)	98 (4.4%)	
Other ^b	7,805 (1.7%)	7,763 (1.7%)	42 (1.9%)	
Assessment centre, n (%)				0.125
England	440,530 (95.8%)	438,358 (95.8%)	2,172 (96.5%)	
Scotland	16,801 (3.7%)	16,729 (3.7%)	72 (3.2%)	
Wales	2,383 (0.5%)	2,377 (0.5%)	6 (0.3%)	
Alcohol status, n (%)				<0.001
Non-drinkers	20,295 (4.4%)	20,150 (4.4%)	145 (6.4%)	
Current drinking	423,128 (92.0%)	421,165 (95.8%)	1,963 (87.2%)	
Previous drinking	16,291 (3.5%)	16,149 (3.5%)	142 (6.3%)	
Smoking status, n (%)				<0.001
Non-smokers	252,005 (54.8%)	251,083 (54.9%)	922 (41.0%)	
Current smoking	47,908 (10.4%)	47,586 (10.4%)	322 (14.3%)	
Previous smoking	159,801 (34.8%)	158,795 (34.7%)	1,006 (44.7%)	
Household income, £, n (%)				<0.001
<18000	89,278 (19.4%)	88,526 (19.4%)	752 (33.4%)	
18000 to 30999	100,349 (21.8%)	99,810 (21.8%)	539 (24.0%)	
31000 to 51999	102,975 (22.4%)	102,624 (22.4%)	351 (15.6%)	
52000 to 100000	80,421 (17.5%)	80,262 (17.5%)	159 (7.1%)	
>100000	21,497 (4.7%)	21,460 (4.7%)	37 (1.6%)	
Other ^c	65,194 (14.2%)	64,782 (14.2%)	412 (18.3%)	
Educational level, n (%)				<0.001
College/university level degree	149,149 (32.4%)	148,650 (32.5%)	499 (22.2%)	
Non-College/university level degree	229,136 (49.8%)	228,122 (49.9%)	1,014 (45.1%)	
Other ^d	81,429 (17.7%)	80,692 (17.6%)	737 (32.8%)	
Diabetes, n (%)				<0.001
Yes	46,128 (10.0%)	45,609 (10.0%)	519 (23.1%)	
No	413,586 (90.0%)	411,855 (90.0%)	1,731 (76.9%)	

^aT-tests and chi-squared tests were used to compare demographic and diabetic characteristics between participants without and with incident pulmonary hypertension, for continuous and categorical variables, respectively. ^bOther here include Chinese, other ethnic group, do not know and prefer not to answer. ^cOther here include do not know and prefer not to answer. ^dOther here include none of the above and prefer not to answer

metabolic pathways that are potentially linked to pulmonary vascular regulation. In contrast, markers of altered energy metabolism including creatinine (HR = 1.120, $P_{FDR} = 8.17 \times 10^{-4}$) and the ketone body metabolite 3-hydroxybutyrate (HR = 1.188, $P_{FDR} = 1.15 \times 10^{-7}$) were associated with significantly higher PH risk. Enhanced

ketogenesis and impaired mitochondrial oxidative function are metabolic abnormalities that have been previously described in patients with PH [30, 31]. Although GlycA (HR = 1.043, $P_{FDR} = 0.247$) and glucose (HR = 1.041, $P_{FDR} = 0.202$) did not reach statistical significance in univariate models, its inclusion in multivariable

Table 2 Associations of diabetes mellitus with risk of incident pulmonary hypertension (n=459,714)

Diabetes	Cases/Person-Years	Model 1		Model 2		Model 3	
		HR (95%CI)	P	HR (95%CI)	P	HR (95%CI)	P
No	1731/5,498,467	1 (Ref.)	-	1 (Ref.)	-	1 (Ref.)	-
Yes	519/591,324	2.191 (1.983,2.420)	<0.001	1.538 (1.384,1.710)	<0.001	1.396 (1.255,1.553)	<0.001

Model 1: adjusted for age, sex and ethnic.
Model 2: adjusted for age, sex, ethnic, BMI, smoking status and drinking status.
Model 3: adjusted for age, sex, ethnic, BMI, smoking status, drinking status, physical activity, income, education, DBP, SBP and assessment centre.

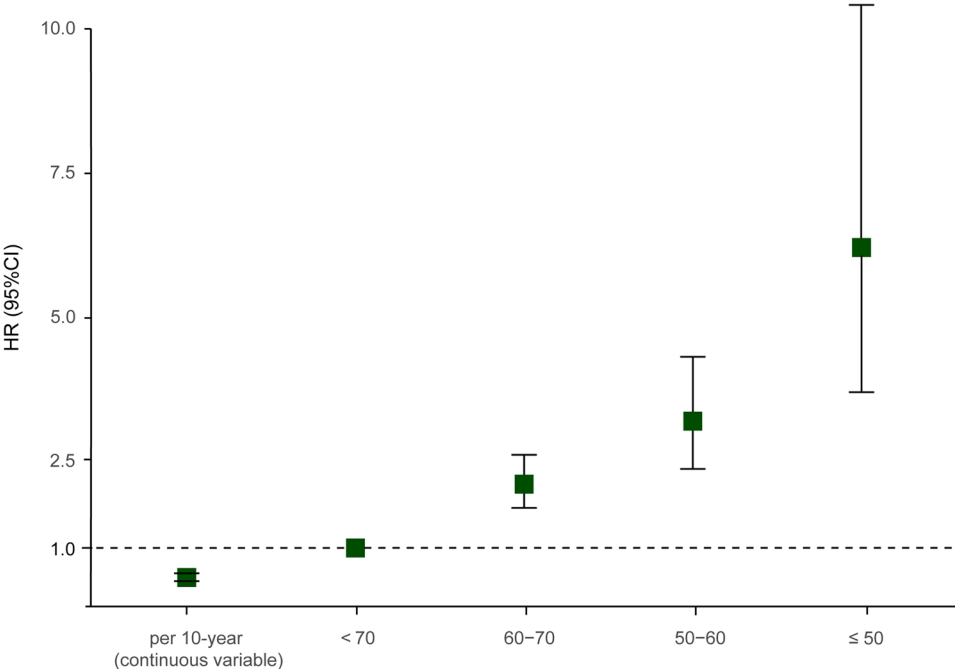


Fig. 2 Association between age at onset of diabetes mellitus and the risk of incident pulmonary hypertension (n = 42,270). For every 10-year increase in the age at onset of DM. Adjusted for age, sex, ethnic, BMI, smoking status, drinking status, physical activity, income, education, DBP, SBP and assessment centre

LASSO models improved predictive performance (C-statistic increased from 0.726 to 0.735). This suggests that GlycA and glucose may act synergistically with other metabolic pathways and contribute to the broader metabolic signature underlying PH risk (Fig. 3B, Table S3).

LASSO modeling identifies seven key metabolites mediating the DM–PH relationship

To identify the most informative metabolites for PH risk prediction, we selected seven metabolites (glucose, GlycA, creatinine, 3-Hydroxybutyrate, VLDL particle mean diameter, leucine, and omega-3 fatty acids) as independent predictors of PH risk (Figure S2). Subsequent mediation analysis revealed that these metabolites collectively explained 37.26% of the total effect of DM on PH risk. Specifically, 3-Hydroxybutyrate, creatinine, GlycA, and glucose exhibited mediating effects of 0.25%, 0.66%, 11.58%, and 31.82%, respectively. Glucose and GlycA were the strongest mediators. Interestingly, VLDL particle mean diameter, leucine, and

omega-3 fatty acids exerted masking effects, attenuating the DM–PH association by 11.46%, 9.53%, and 3.23%, respectively (Fig. 4, Table S4).

Enhanced predictive performance of the model with metabolites

To evaluate the clinical utility of metabolomic profiling, we assessed whether the inclusion of selected metabolites could improve PH risk prediction beyond traditional clinical models. The basic model, which included demographic, lifestyle, and comorbidities, yielded a C-statistic of 0.692 (95% CI: 0.660–0.724). While the incorporation of seven key metabolites significantly improved this discriminative ability, increasing the C-statistic to 0.735 (95% CI: 0.704–0.766, *P* < 0.0001) (Table 3). And significant gains in NRI and IDI metrics, with an NRI of 0.294 (95% CI: 0.044–0.309, *P* < 0.001) and an IDI of 0.004 (95% CI: 0.001–0.005, *P* < 0.001), confirming that the metabolites contribute substantial incremental predictive value.

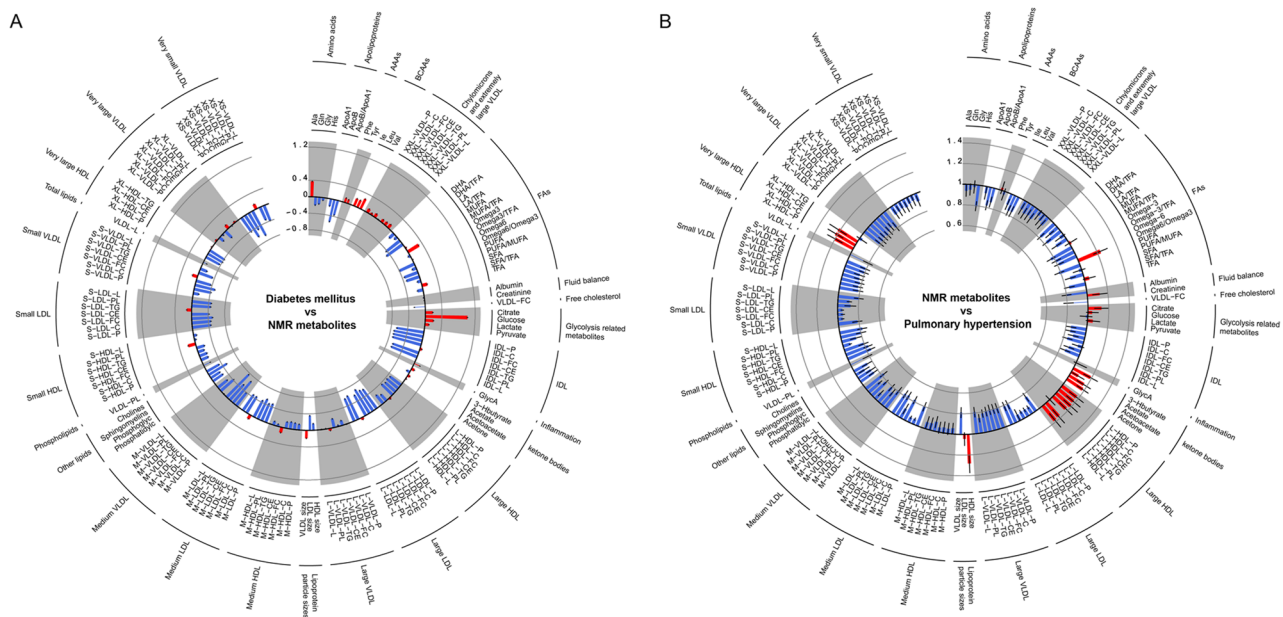


Fig. 3 (A) Associations between diabetes mellitus and 143 NMR-derived metabolites ($n = 237,848$). (B) Associations between 143 NMR-derived metabolites and the risk of incident pulmonary hypertension ($n = 237,848$). For Panel A, bars pointing to the right (red) indicate metabolites higher in participants with DM, and bars pointing to the left (blue) indicate metabolites lower in DM. For Panel B, red bars indicate metabolites for which higher levels are associated with increased PH risk (HR > 1 per 1-SD), and blue bars indicate metabolites for which higher levels are associated with reduced PH risk (HR < 1 per 1-SD). Metabolites are organized clockwise into biologically meaningful classes, starting from amino acids, apolipoproteins, essential amino acids, branched-chain amino acids, fatty acids, fluid balance related metabolites, glycolysis-related metabolites, inflammation, ketone bodies, lipid subclasses across VLDL/DL/LDL/HDL and lipoprotein particle sizes. To aid biological interpretation, we categorized the key metabolites into major functional classes—lipid metabolism, energy and mitochondrial metabolism, inflammation, amino-acid metabolism, and glucose (glycolytic activity). We used these categories when interpreting mediation and prediction models. All models were adjusted for age, sex, ethnicity, BMI, smoking status, drinking status, physical activity, household income, education, DBP, SBP, and assessment center

Stratified and sensitivity analyses confirm the robustness of findings

We performed stratified analyses across sex, age, BMI, drinking intake status, smoking status, physical activity, household income, educational level, and SBP and DBP. The significant positive associations between DM and PH remained consistent across most subgroups, but diminished among participants with household incomes exceeding £100,000 and for previous or never drinkers, possibly due to insufficient statistical power as the number of incidence cases were all <150 individuals. Significant interactions were observed with age, BMI, and household income ($P < 0.05$). Notably, the association between DM and PH was stronger in younger individuals, those with obesity, and those from higher-income backgrounds (Fig. 5). In sensitivity analyses, the robustness of the findings was consistently demonstrated across multiple scenarios (Table S5).

Discussion

This study provides robust epidemiological evidence that DM is an independent risk factor for PH, overcoming limitations of prior small-scale studies. Using a large, diverse cohort with long follow-up, we show that DM markedly increases PH risk, especially with earlier onset,

likely reflecting prolonged metabolic stress. This aligns with the UK and Irish PH registry data, which reported a DM prevalence exceeding 20% in pulmonary arterial hypertension (PAH) patients aged ≥ 50 years [32]. Our study extends these by linking early-onset DM to worse hemodynamic profiles, underscoring the value of early screening and intervention in younger diabetes.

Current therapies for PH primarily aim to relieve pulmonary vasoconstriction, but their clinical benefits are often limited in patients with comorbid DM [33]. Our findings indicate that glucose and GlycA mediate a substantial proportion of the DM–PH association, highlighting key metabolic pathways involved. These results are consistent with recent experimental evidence showing that augmented endothelial glycolysis in the diabetic lung accelerates pulmonary vascular remodeling and promotes the development of severe PH. Moreover, in hypoxia-induced PH models in diabetic mice, pharmacological inhibition of glycolysis alleviates disease progression, further reinforcing the role of glucose metabolism and glycolytic intermediates [34].

Beyond endothelial dysfunction, accumulating evidence indicates that chronic hyperglycemia induces a glycolytic shift in PSMCs, leading to the accumulation of glycolytic intermediates that promote proliferation,

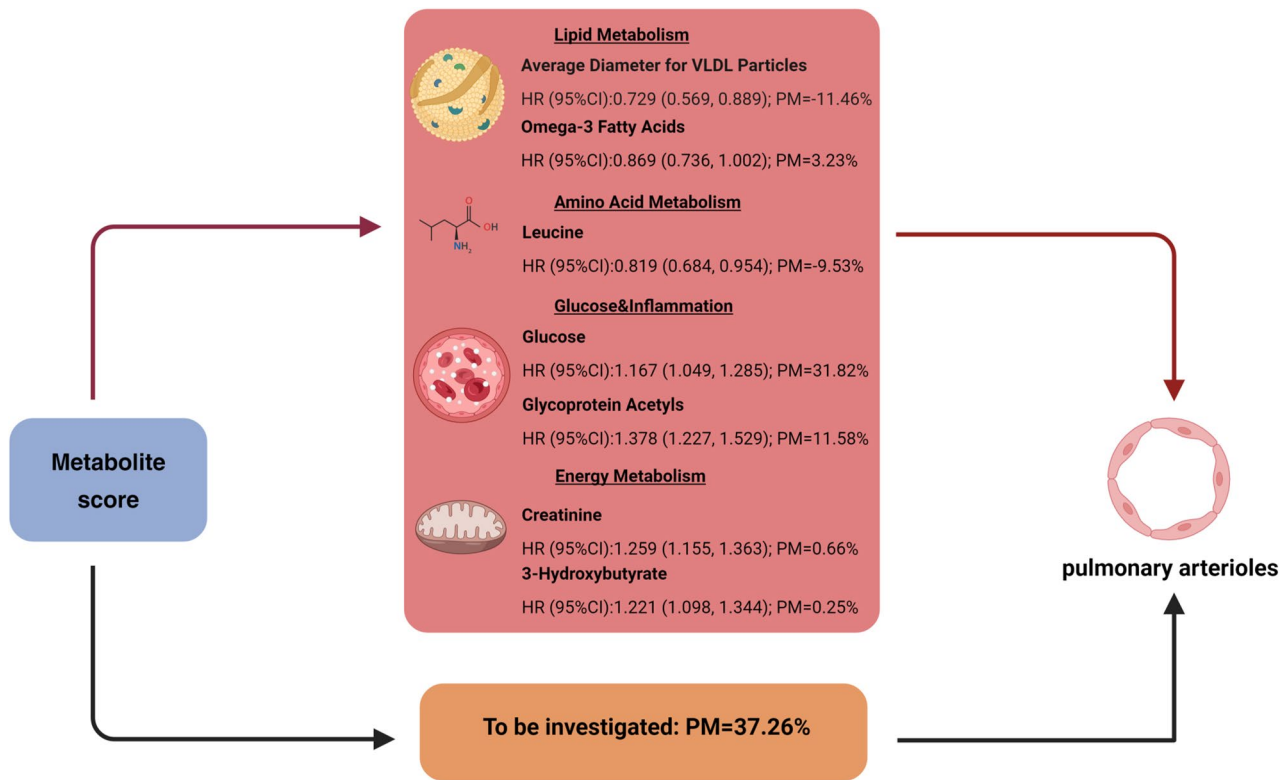


Fig. 4 Mediated effects by 7 NMR metabolites on the association of Diabetes mellitus with Risk of Incident Pulmonary hypertension ($n = 237,848$). Metabolites were grouped into four biological categories based on their primary physiological roles: (1) Lipid Metabolism (average VLDL particle diameter, omega-3 fatty acids); (2) Amino Acid Metabolism (leucine); (3) Glucose & Inflammation (glucose, GlycA); and (4) Energy Metabolism (creatinine, 3-hydroxybutyrate). This categorization highlights the major metabolic domains through which DM may influence pulmonary vascular remodeling. The percent mediated (PM) estimates represent the proportion of the association between DM and PH mediated by each metabolite. Adjusted for age, sex, ethnic, BMI, smoking status, drinking status, physical activity, income, education, DBP, SBP and assessment centre

Table 3 Performance of risk prediction models for 10-year risk of incident pulmonary hypertension by including LASSO-selected metabolites ($n=23,478$)

10-year risk of pulmonary hypertension		Basic model ^a	Basic model + 7 metabolites ^b
C Statistic	Estimates	0.692 (0.660,0.724)	0.735 (0.704,0.766)
	P for difference	Reference	<0.0001
Continuous NRI		Reference	0.294 (0.044,0.309)
Absolute IDI		Reference	0.004 (0.001,0.005)

^aBasic model: age, sex, ethnic, BMI, smoking status, drinking status, physical activity, income, education, DBP, SBP and assessment centre, ^bThe 7 metabolic measures remained significant after LASSO selection and mutual adjustment: Average Diameter for VLDL Particles, Leucine, Omega-3 fatty acids, 3-Hydroxybutyrate, Creatinine, Glycoprotein Acetyls, Glucose.

apoptosis resistance, and extracellular acidosis [35]. The resulting acidotic microenvironment activates pH-sensitive ion channels such as ASIC1a, further amplifying downstream inflammatory signaling [36]. In parallel, persistent hyperglycemia increases the formation of advanced glycation end product (AGEs), which activates the AGE-RAGE axis and promote upregulation of HIF- α and the NLRP3 inflammasome [35]. These changes intensify oxidative stress and low-grade systemic

inflammation, the latter reflected by elevated circulating GlycA [37, 38]. These metabolic and inflammatory disturbances converge to drive vascular remodeling [39]. Experimental inhibition of HIF- α , NLRP3, or RAGE mitigates vascular injury in diabetic models [39, 40], while clinical data link hyperglycemia-related inflammation with adverse PAH outcomes [41, 42]. Collectively, these findings support a gluco-inflammatory model explaining poor vasodilator efficacy in diabetic PH.

Lipid metabolism also appears to play a critical role in pulmonary vascular remodeling. Preclinical studies have shown that metformin and pioglitazone attenuate pulmonary vascular remodeling through activation of PPAR γ , a nuclear receptor essential for lipid homeostasis. PPAR γ agonists enhance mitochondrial fatty acid oxidation and ATP generation, counteracting the mitochondrial inefficiency observed in PAH [43–45]. Other lipid-related abnormalities, such as reduced plasma lipocalin levels and decreased apolipoprotein expression, have been implicated in the development of PAH in animal models [46]. Consistent with these mechanistic insights, our findings reveal broad lipid-related alterations associated with PH, extending from lipoprotein particle characteristics

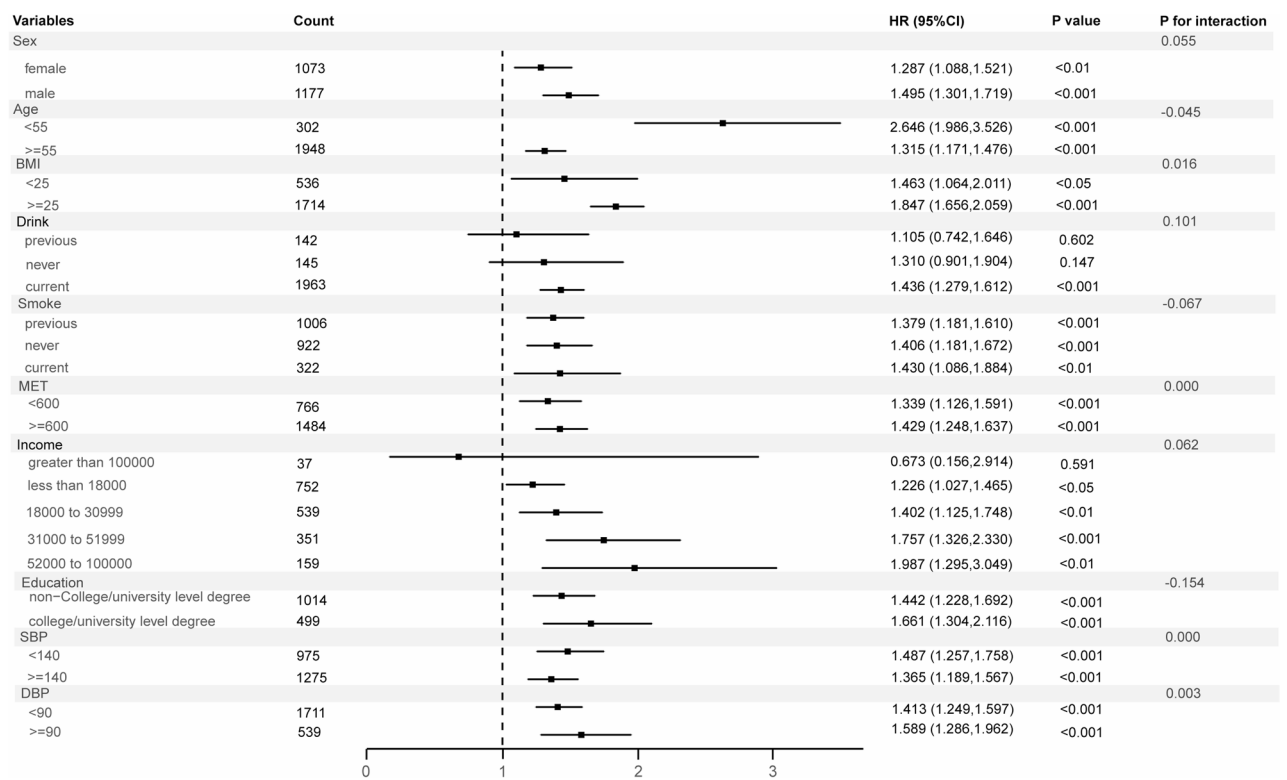


Fig. 5 Stratified analyses of multiple demographic and lifestyle factors. Factors included gender, age, body mass index, alcohol use, smoking, physical activity, household income, education, SBP and DBP

to fatty-acid profiles. Enlarged VLDL, which carry more triglycerides than smaller VLDL subclasses, has been associated to endothelial stress and vascular dysfunction [27, 28]. Conversely, omega-3 fatty acids, including EPA and DHA, act as a metabolic counterbalance against the adverse effects of triglyceride-rich VLDL particles by activating anti-inflammatory and vascular protective pathways [47]. They exert direct anti-inflammatory and vascular protective effects by activating the NO/sGC/cGMP pathway and inhibiting the production of pro-inflammatory and pro-fibrotic lipid mediators such as prostaglandins, leukotrienes, and thromboxanes. Furthermore, omega-3 fatty acids effectively reduce the expression of key pro-inflammatory cytokines such as TNF- α , IL-1 β , IL-6, and IL-8, which serve as key mediators of inflammatory responses and vascular remodeling in the pathophysiology of PAH [48, 49]. These mechanistic actions support the potential that omega-3 fatty acids as an adjunct to current PH therapies. Growing preclinical evidence also suggests they could represent a promising long-term nutritional intervention for PAH [50].

Energy metabolism may exert dual effects. The ketone body 3-hydroxybutyrate showed risk-enhancing in our diabetes cohort. However, experimental studies in non-diabetes models report vasodilatory and right ventricular energetics-enhancing effects of β -hydroxybutyrate [31]. The emerging view that low concentrations of

ketone bodies, particularly β -hydroxybutyrate, may help improve endothelial and vascular function in metabolic diseases. Conversely, the high concentrations of ketone body observed in diabetic ketoacidosis contribute to diabetic vascular disease and vascular complications [51]. Moreover, elevated creatinine was associated with higher PH risk, possibly reflecting renal hypoperfusion secondary to right-ventricular dysfunction [52], underscoring the detrimental impact of cardiorenal axis impairment on PH progression. Amino-acid metabolism may also be relevant to PH. Research suggests that leucine enhances insulin sensitivity and promotes both glucose and lipid metabolism [53]. This may be due to leucine metabolites activating the mTOR signaling pathway, significantly increasing leptin secretion from adipose tissue and subsequently suppressing lipogenesis [54, 55]. These metabolic effects plausibly explain the inverse association between circulating leucine levels and PH risk observed in our study.

These observations highlight the complex interplay between lipid, inflammation, amino-acid, and energy metabolism in PH, emphasizing the potential of metabolic modulation as a therapeutic strategy. Future studies are needed to define the molecular pathways underlying these associations and to evaluate whether targeted metabolic interventions can improve clinical outcomes in PH.

Finally, risk assessment is essential for optimizing the management of PH [56]. Current ESC guideline-based models underestimate risk in patients with comorbid DM [57], who show poorer treatment responses and outcomes [25]. Incorporating metabolomic data improved prediction by capturing subclinical alterations. This enhanced precision in identifying high-risk patients has the potential to guide personalized management strategies, including the optimization of pharmacologic regimens based on individual metabolic signatures.

Limitations

There are still some limitations to consider. Firstly, the reliance on single-timepoint metabolite quantification at baseline limits our ability to capture dynamic metabolic shifts and temporal variations that may modulate PH progression. Secondly, while extensive covariate adjustments were implemented, the potential influence of unmeasured or residual confounding factors cannot be fully excluded. Thirdly, our classification of PH using ICD-10 codes is consistent with large-scale epidemiological studies, but further stratification by clinical subtype (e.g., group 1 PAH versus group 2/3 PH) may refine the specificity of metabolic pathways in the cohort [57]. Finally, despite rigorous internal validation through cross-validation and iterative assessment, external validation in ethnically and geographically diverse populations has not been undertaken and further research is needed for the future.

Conclusions

In conclusion, our study demonstrates that DM elevates PH risk, partially mediated by metabolic dysregulation involving lipid remodeling, hyperglycemia, inflammation, BCAA accumulation, and ketone elevation. Our metabolomic prediction model is expected to enhance early identification and prevention in high-risk populations.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12931-026-03499-4>.

Supplementary Material 1.

Acknowledgements

We thank all UK Biobank participants and everyone involved in building the UK Biobank study.

Clinical trial registration

This work is an observational cohort study and does not involve an interventional clinical trial. Clinical trial registration number: Not applicable.

Authors' contributions

X.C., L.Y., and M.Y. contributed to the study concept and design. L.Y. provided statistical expertise. X.C. conducted data charting and visualization. X.C., W.L., Y.H., J.Z., and Z.F. performed the statistical analyses. X.C. and L.Y. drafted the manuscript. X.W., D.J., and X.Y. reviewed the manuscript and verified the

accuracy of the data. All authors reviewed and approved the final version of the manuscript for submission.

Funding

This work was supported by the National Natural Science Foundation of China (Grant Nos. 82370414) and the Hubei Provincial Natural Science Foundation of China (Grant Nos. 2021CFB421, 2022CFB250).

Data availability

The usage of UK Biobank data has been approved by UK Biobank Research Team (Application ID: 106528). Data from UK Biobank (URL: <https://www.ukbiobank.ac.uk/>) are available on application.

Declarations

Ethics approval and consent to participate

This study was conducted in accordance with the principles of the World Medical Association Declaration of Helsinki. UK Biobank received ethical approval from the Northwest Multicenter Research Ethics Committee, and all participants signed informed consent forms at enrolment.

Consent for publication

Because the dataset was anonymized and no individual-level identifiable information (including images or personal clinical details) is presented in this manuscript, consent for publication is not applicable.

Competing interests

The authors declare no competing interests.

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Received: 24 August 2025 / Accepted: 4 January 2026

Published online: 19 January 2026

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